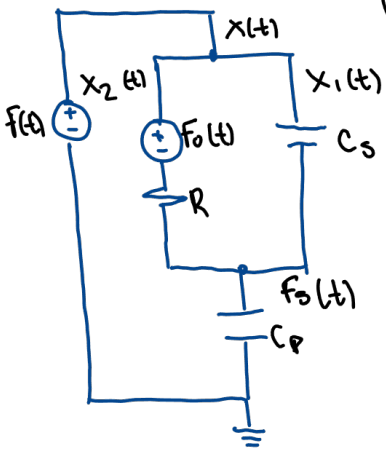
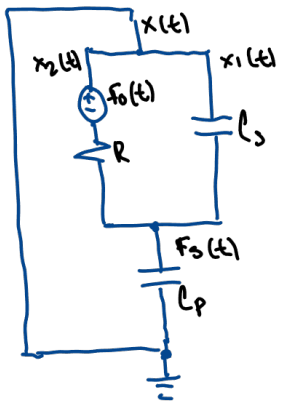
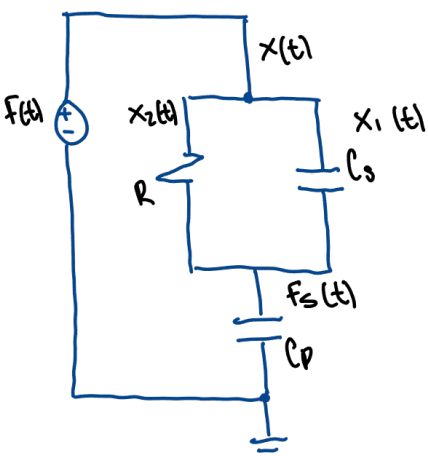
Practica 3: Sistema musculoesqueletico

(circuito original).



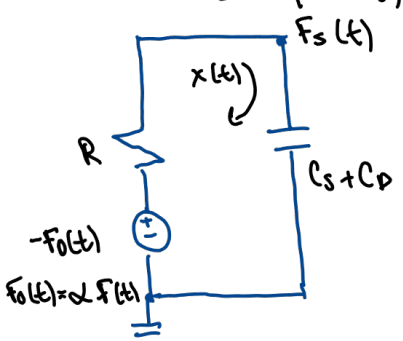
Circuito 2





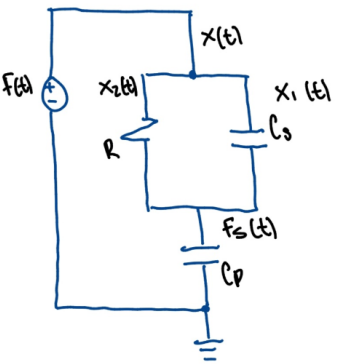
Circuito 1

(Circuito 2 simplificado).



Analisis del circuito por superposición

Circuito 1 (apagando $f_0(t)$)



$$x(t) = x_1(t) + x_2(t)$$

$$x(t) = C_p \frac{df_{s1}(t)}{dt}$$

$$x_1(t) = C_s \frac{d[F(t) - f_{s1}(t)]}{dt}$$

$$x_2(t) = \frac{F(t) - f_{s1}(t)}{R}$$

$$C_p \frac{df_{s1}(t)}{dt} = C_s \frac{d[F(t) - f_{s1}(t)]}{dt} + \frac{F(t) - f_{s1}(t)}{R}$$

$$C_p s F_{s1}(s) = C_s s [F(s) - F_{s1}(s)] + \frac{F(s) - F_{s1}(s)}{R}$$

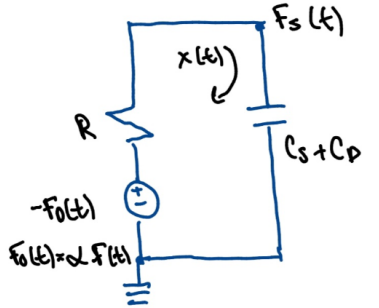
$$C_p s F_{s1}(s) = C_s s F(s) - C_s s F_{s1}(s) + \frac{F(s)}{R} - \frac{F_{s1}(s)}{R}$$

$$\left(C_p s + C_s s + \frac{1}{R}\right) F_{s1}(s) = \left(C_s s F(s) + \frac{1}{R}\right) F(s)$$

$$\frac{R(C_p + C_s)s + 1}{R} F_{s1}(s) = \frac{C_s R s + 1}{R} F(s)$$

$$f_{s1}(s) = \frac{\frac{C_s R s + 1}{R} F(s)}{\frac{R(C_p + C_s)s + 1}{R}} = \frac{C_s R s + 1}{R(C_p + C_s)s + 1} F(s)$$

(Circuito 2 simplificado).



Circuito 2 (apagando F(t))

$$-\alpha F(t) = R x(t) + \frac{1}{C_P + C_S} \int x(t)\,dt$$

$$F_{S_2}(t) = \frac{1}{C_P + C_S} \int x(t)\,dt$$

$$-\alpha F(s) = R X(s) + \frac{X(s)}{(C_P + C_S)s}$$

$$F_{S_2}(s) = \frac{X(s)}{(C_P + C_S)s}$$

$$F(s) = -\frac{X(s)}{\alpha} \frac{R(C_P + C_S)s + 1}{(C_P + C_S)s}$$

$$\frac{F_{S_2}(s)}{F(s)} = \frac{\frac{X(s)}{(C_P + C_S)s}}{-\frac{X(s)}{\alpha} \frac{R(C_P + C_S)s + 1}{(C_P + C_S)s}}$$

$$F_{S_2}(s) = -\frac{\alpha F(s)}{R(C_P + C_S)s + 1}$$

Función de transferencia

$$F_S(s) = F_{S_1}(s) + F_{S_2}(s) = \frac{C_S R_S + 1}{R(C_P + C_S)s + 1}$$

$$F_S(s) = \frac{C_S R_S + 1 - \alpha}{R(C_P + C_S)s + 1} F(s)$$

$$\frac{F_S(s)}{F(s)} = \frac{C_S R_S + 1 - \alpha}{R(C_P + C_S)s + 1}$$